

Machine Learning-Lecture 5

P1.py

```
C:\ml\L5>python p1.py
```

```
place area
```

```
0 Lonavala 2600
```

```
1 Lonavala 3000
```

```
2 Lonavala 3200
```

```
3 Karjat 2600
```

```
4 Karjat 2800
```

```
5 Khandala 2600
```

```
6 Khandala 2900
```

```
7 Khandala 3200
```

```
enter area 3000
```

```
1.karjat , 2.Khandala ,3.Lonavla 3
```

```
[582931.03448276]
```

P2.py

```
C:\ml\L5>python p2.py
```

```
state city area
```

```
0 mah mumbai 1000
```

```
1 mah mumbai 1200
```

```
2 mah pune 1000
```

```
3 mah pune 800
```

```
4 guj surat 1200
```

```
5 guj surat 5000
```

```
6 guj amd 2000
```

```
7 guj amd 2300
```

```
area state_guj ... city_pune city_surat
```

```
0 1000 False ... False False
```

1	1200	False	...	False	False
2	1000	False	...	True	False
3	800	False	...	True	False
4	1200	True	...	False	True
5	5000	True	...	False	True
6	2000	True	...	False	False
7	2300	True	...	False	False

[8 rows x 7 columns]

P3.py

C:\ml\L5>python p3.py

```

state  city area price
0 mah  mumbai 1000  200
1 mah  mumbai 1200  210
2 mah  pune   1000  45
3 mah  pune   800   35
4 guj  surat  1200  60
5 guj  surat  5000  200
6 guj  amd    2000  120
7 guj  amd    2300  150

state  city
0 mah  mumbai
1 mah  mumbai
2 mah  pune
3 mah  pune
4 guj  surat
5 guj  surat
6 guj  amd
7 guj  amd

area
0 1000

```

1 1200

2 1000

3 800

4 1200

5 5000

6 2000

7 2300

[[0. 1. 0. 1. 0. 0.]

[0. 1. 0. 1. 0. 0.]

[0. 1. 0. 0. 1. 0.]

[0. 1. 0. 0. 1. 0.]

[1. 0. 0. 0. 0. 1.]

[1. 0. 0. 0. 0. 1.]

[1. 0. 1. 0. 0. 0.]

[1. 0. 1. 0. 0. 0.]]

state_guj state_mah ... city_pune city_surat

0 0.0 1.0 ... 0.0 0.0

1 0.0 1.0 ... 0.0 0.0

2 0.0 1.0 ... 1.0 0.0

3 0.0 1.0 ... 1.0 0.0

4 1.0 0.0 ... 0.0 1.0

5 1.0 0.0 ... 0.0 1.0

6 1.0 0.0 ... 0.0 0.0

7 1.0 0.0 ... 0.0 0.0

[8 rows x 6 columns]

area state_guj ... city_pune city_surat

0 1000 0.0 ... 0.0 0.0

1 1200 0.0 ... 0.0 0.0

2 1000 0.0 ... 1.0 0.0

3 800 0.0 ... 1.0 0.0

4 1200 1.0 ... 0.0 1.0

5 5000 1.0 ... 0.0 1.0

```
6 2000    1.0 ...    0.0    0.0
7 2300    1.0 ...    0.0    0.0

[8 rows x 7 columns]
```

P4.py

```
C:\ml\L5>python p4.py
```

```
Level
```

```
0  1
1  2
2  3
3  4
4  5
5  6
6  7
7  8
8  9
9 10
```

```
[[1.000e+00 1.000e+00 1.000e+00 1.000e+00 1.000e+00]
 [1.000e+00 2.000e+00 4.000e+00 8.000e+00 1.600e+01]
 [1.000e+00 3.000e+00 9.000e+00 2.700e+01 8.100e+01]
 [1.000e+00 4.000e+00 1.600e+01 6.400e+01 2.560e+02]
 [1.000e+00 5.000e+00 2.500e+01 1.250e+02 6.250e+02]
 [1.000e+00 6.000e+00 3.600e+01 2.160e+02 1.296e+03]
 [1.000e+00 7.000e+00 4.900e+01 3.430e+02 2.401e+03]
 [1.000e+00 8.000e+00 6.400e+01 5.120e+02 4.096e+03]
 [1.000e+00 9.000e+00 8.100e+01 7.290e+02 6.561e+03]
 [1.000e+00 1.000e+01 1.000e+02 1.000e+03 1.000e+04]]
```

```
enter level 3
```

```
[58642.1911421]
```

P4a.py

